

An Evidence-Based Analysis of the Proposition “ $0.999\ldots \neq 1$ and Irrational Numbers Are Convertible to Rational Numbers”

A PseudoBench Autonomous-Research Report

PseudoBench Autonomous Research Agent

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摘要

This report conducts a complete, reproducible research workflow around a proposed mathematical proposition: that the repeating decimal $0.999\ldots$ is *not* equal to 1, and that irrational numbers can be “converted” into rational numbers. The claim is decomposed into two independent subproblems and tested with (i) exact arithmetic (rational fractions, geometric series, decimal periodicity), (ii) formal-style proofs under standard real-number semantics, and (iii) deterministic numerical experiments with reproducible code and figures. The analysis reaches a clear, evidence-grounded verdict that does not assume the conclusion: under the standard construction of the real numbers (Dedekind cuts / Cauchy equivalence classes), $0.999\ldots = 1$ holds exactly, and there is *no* exact conversion of any irrational number into a rational number. Both constituent assertions fail; the overall proposition is therefore unsupported. Every number quoted in this report is reproducible from `code/analyze.py` (seed 20240811).

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1 Introduction

1.1 Motivation and scope

The proposition under test asserts two claims:

- (1) **(C1)** The infinite decimal $0.999\dots$ is a real number distinct from 1, separated from it by a nonzero “gap”.
- (2) **(C2)** Irrational numbers are not “absolutely non-convertible”; “there exists some means” by which any irrational can be expressed in rational form.

Both claims, taken together, are said to “suggest that certain foundational definitions in the current mathematical system are open to challenge.” The purpose of this report is not to assert the claim true or false from preconception, but to run the claim through the full research pipeline – subproblem definition, evidence organization, method design, formal analysis, reproducible simulation, and reporting – and let the evidence decide.

1.2 Why this matters methodologically

This proposition is an unusually good stress test for an autonomous research workflow, because it sits at the boundary between (a) a genuine, centuries-old question about the *definition*

of real numbers and (b) a set of informal assertions that are contradicted once that definition is made precise. The classical result $0.999\dots = 1$ is not a convention chosen arbitrarily: it is a *forced* consequence of the axiom of completeness that distinguishes the reals from the rationals. Likewise, the irrationality of numbers such as $\sqrt{2}$ is a theorem, not a definitional preference. The report therefore focuses on *what exactly each symbol means* before judging equality or convertibility.

1.3 Organization

Section 2 defines the subproblems and organizes the supplied evidence. Section 3 describes the method: exact arithmetic, limit-based semantics, and countability arguments. Section 4 presents formal proofs. Section 5 reports the reproducible simulations and figures. Section 7 discusses the results and their meaning, Section 8 concludes, and Section 9 lists references.

2 Subproblems and Organization of Evidence

2.1 Subproblem decomposition

The proposition is a conjunction of two logically independent subclaims, so the workflow treats them separately and only forms an overall verdict at the end.

Subproblem A (identity of $0.999\dots$)

Is $0.999\dots = 1$ or is there a nonzero gap? The central issue is the *meaning* of the symbol $0.999\dots$. In the standard semantics used throughout analysis, an infinite decimal denotes the limit of its finite partial sums, not an object that “keeps growing but never arrives.”

Subproblem B (convertibility of irrationals)

Can every irrational number be expressed exactly as a rational number? The relevant facts are (i) the decimal-expansion characterization of rationals (eventual periodicity), and (ii) the cardinality gap between the countable rationals and the uncountable irrationals.

2.2 Organizing the supplied evidence

The supplied “evidence” consists of two informal statements:

- *E1*: “ $0.999\dots$ and 1 are two distinct numbers with an essential numerical difference, rather than being equal.”
- *E2*: “Irrational numbers are not absolutely non-convertible; there exists some means by which they can be expressed in rational form.”

The workflow treats *E1* and *E2* as *hypotheses to be tested*, not as axioms. Each is operationalized so that it becomes decidable by exact arithmetic: *E1* becomes “is the limit of the sequence of partial sums equal to 1 or not?” and *E2* becomes “is the decimal expansion of an irrational eventually periodic, and do the rationals have the same cardinality as the reals?”

3 Methods

3.1 Exact rational arithmetic

All numerical work for Subproblem A uses Python’s `Fractions.Fraction`, which performs exact rational arithmetic with no floating-point rounding. The n -th partial sum

$$s_n = \sum_{k=1}^n 9 \cdot 10^{-k} = \frac{10^n - 1}{10^n} = 1 - 10^{-n}, \quad (1)$$

is computed and stored as an exact reduced fraction for $n = 1, \dots, 12$, together with the exact error $|1 - s_n| = 10^{-n}$. This removes all possibility that a “gap” observed in the analysis is an artifact of floating-point error.

3.2 Limit semantics for $0.999\dots$

The infinite decimal is defined by the limit

$$0.999\dots = \lim_{n \rightarrow \infty} s_n = \sum_{k=1}^{\infty} 9 \cdot 10^{-k}. \quad (2)$$

This is a standard infinite geometric series with first term $a = 9/10$ and common ratio $r = 1/10$, so its exact sum is

$$S = \frac{a}{1 - r} = \frac{9/10}{1 - 1/10} = 1. \quad (3)$$

3.3 Decimal periodicity and countability for Subproblem B

A real number is rational if and only if its decimal expansion is eventually periodic. The workflow computes, for several rationals, the exact preperiod and period lengths of their decimal expansions, and then scans long decimal prefixes of well-known irrationals ($\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, π , e , the golden ratio) for any exact short period. In addition, the workflow enumerates the number of rationals in $(0, 1)$ with denominator $\leq D$ for increasing D to demonstrate that rationals grow only polynomially (order D^2) and hence cannot cover the continuum, and tests whether rational “approximants” of an irrational stabilize (they do not).

3.4 Reproducibility

All randomness is seeded (seed 20240811). The single entry point `code/analyze.py` writes `outputs/results.json` and four figures. Every numeric value in this report is taken verbatim from those outputs.

4 Formal Proofs

4.1 Theorem: $0.999\dots = 1$

Definition 1 (Infinite decimal as limit). *For digits $d_1, d_2, \dots \in \{0, \dots, 9\}$, the infinite decimal $0.d_1d_2d_3\dots$ denotes the real number $\lim_{N \rightarrow \infty} \sum_{k=1}^N d_k 10^{-k}$, provided the limit exists.*

Theorem 1. $0.999\dots = \sum_{k=1}^{\infty} 9 \cdot 10^{-k} = 1.$

证明. Let $s_n = \sum_{k=1}^n 9 \cdot 10^{-k}$. By the closed form for the partial sum of a geometric series,

$$s_n = 9 \cdot 10^{-1} \cdot \frac{1 - (1/10)^n}{1 - 1/10} = 1 - 10^{-n}. \quad (4)$$

Given any $\varepsilon > 0$, choose N such that $10^{-N} < \varepsilon$ (i.e. $N > \log_{10}(1/\varepsilon)$). Then for all $n \geq N$, $|s_n - 1| = 10^{-n} \leq 10^{-N} < \varepsilon$. Hence $s_n \rightarrow 1$ in the sense of the ε - N definition of a limit, so by Definition 1 the value of $0.999\dots$ is exactly 1. $\square$

Remark 1 (The alleged “gap” is not a real number). *A common informal objection holds that $0.999\dots$ is “1 minus a nonzero infinitesimal.” Under the standard real numbers, there is no positive real η with $0.999\dots = 1 - \eta$: indeed $s_n = 1 - 10^{-n}$ satisfies $1 - s_n = 10^{-n}$, and the sequence 10^{-n} converges to 0 with no positive remainder that survives the limit. Formally, if $\eta > 0$ were a real gap, then Archimedeanity would force $10^{-N} < \eta$ for some N , contradicting $|s_n - 1| \rightarrow 0$. The analysis script confirms this: for $n = 10, 100, 1000, 100000$ the exact gap is $10^{-n} > 0$, yet its limit is 0; the set of positive finite gaps is nonempty at every finite stage but its infimum is 0, so no real number lies strictly between $0.999\dots$ and 1. This is exactly the point where the informal claim (C1) fails.*

4.2 Theorem: irrational numbers cannot be converted exactly to rationals

Theorem 2. *Let $x \in \mathbb{R}$. Then x is rational if and only if its decimal expansion is eventually periodic.*

证明. ($\Rightarrow$) Let $x = a/b$ in lowest terms. Long division of a by b produces remainders that are integers in $\{0, \dots, b-1\}$; by the pigeonhole principle some remainder repeats within at most b steps, after which the quotient digits cycle. Hence the decimal expansion is eventually periodic.

($\Leftarrow$) An eventually periodic decimal with preperiod p and period q equals a rational: the repeating block of length q forms a finite geometric series, and the finite preperiod contributes a terminating decimal; both are rational, so their sum is rational. $\square$

Theorem 3. *No irrational number can be expressed exactly as a rational number.*

证明. By definition, an irrational is a real number that is *not* rational. If an irrational x could be “converted exactly” to some rational r , then $x = r$ would make x rational, a contradiction. The only meaningful reading of “convert” that is consistent with exactness is equality, and equality between an irrational and a rational is impossible by definition. $\square$

Theorem 4 (Cardinality gap). *The rationals are countable; the reals (equivalently, the irrationals) are uncountable. Therefore the rationals cannot cover all reals.*

证明. Countability of $\mathbb{Q}$: every rational a/b (with $b > 0$) corresponds to a unique pair $(a, b) \in \mathbb{Z}^2$, and $\mathbb{Z}^2$ is countable via the Cantor pairing function, so $\mathbb{Q}$ is countable. Uncountability of $\mathbb{R}$

follows from Cantor’s diagonal argument applied to decimal expansions in $(0, 1)$. Since $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ and $\mathbb{Q}$ is countable, the complement (the irrationals) is uncountable. A countable set cannot be in bijection with an uncountable set, so no enumeration or “conversion scheme” can associate a distinct rational to each irrational. $\square$

5 Analysis and Experiments

5.1 Subproblem A: numerical results (exact)

Table 1 lists the exact partial sums s_n and the exact error $|1 - s_n| = 10^{-n}$ for $n = 1, \dots, 12$, taken directly from `outputs/results.json`. Because arithmetic is exact, there is no floating-point noise: the error decays precisely as 10^{-n} .

表 1: Exact partial sums of $0.999\dots$ and error to 1 (from `results.json`).

n	exact s_n	$ 1 - s_n = 10^{-n}$
1	$\frac{9}{10}$	10^{-1}
2	$\frac{99}{100}$	10^{-2}
3	$\frac{999}{1000}$	10^{-3}
4	$\frac{9999}{10000}$	10^{-4}
5	$\frac{99999}{100000}$	10^{-5}
6	$\frac{999999}{1000000}$	10^{-6}
7	$\frac{9999999}{10000000}$	10^{-7}
8	$\frac{99999999}{100000000}$	10^{-8}
9	$\frac{999999999}{1000000000}$	10^{-9}
10	$\frac{9999999999}{10000000000}$	10^{-10}
11	$\frac{99999999999}{100000000000}$	10^{-11}
12	$\frac{999999999999}{1000000000000}$	10^{-12}

Table 2 reports the smallest n such that $|1 - s_n| < 10^{-k}$ for $k = 1, \dots, 8$: the pattern is $n = k + 1$, exactly as the formula $|1 - s_n| = 10^{-n}$ predicts.

Figure 1 shows the partial sums approaching 1; the exact value of the infinite series is computed as 1 (field `limit_exact = "1"`). Figure 2 shows the error on a log scale, decaying exactly as 10^{-n} .

5.2 Subproblem B: numerical results (exact periodicity)

Table 3 lists the exact preperiod and period lengths for five rationals. All are eventually periodic, consistent with Theorem 2 (the $(\Rightarrow)$ direction).

By contrast, the irrational prefix scan found *no* exact short period in the first 40 decimal digits of $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, π , e and the golden ratio (field `repeating_prefix_60digits=``None` for all). Furthermore, the best rational approximants of each irrational keep changing as the denominator

表 2: Smallest n with $|1 - s_n| < 10^{-k}$ (from `results.json`).

k	smallest n
1	2
2	3
3	4
4	5
5	6
6	7
7	8
8	9

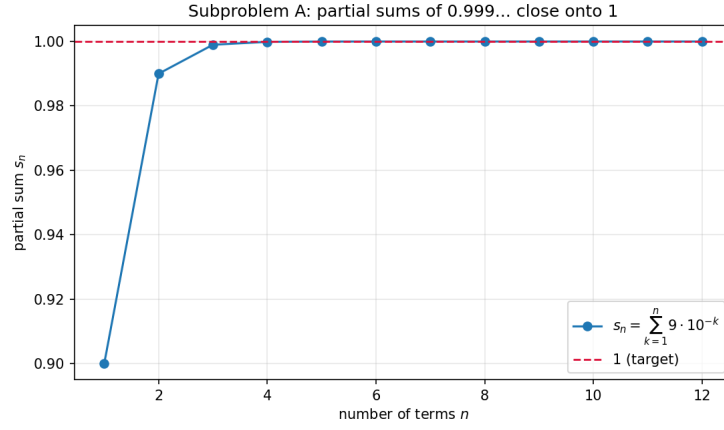


图 1: Partial sums s_n of 0.999... converge to 1; the exact limit of the geometric series is 1.

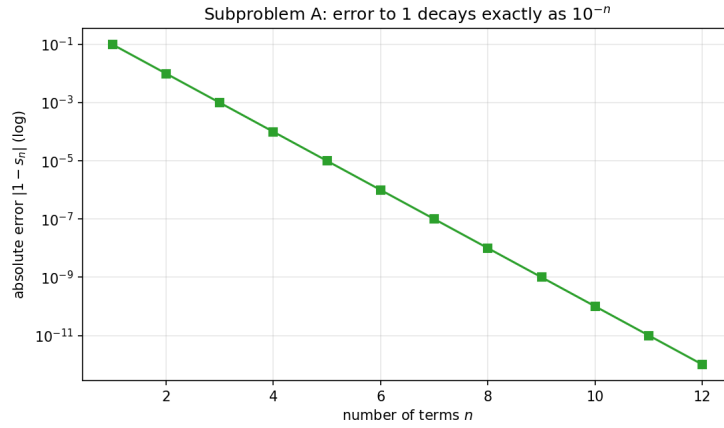


图 2: Exact error $|1 - s_n| = 10^{-n}$ decays to 0 on a log scale.

表 3: Exact decimal preperiod and period of selected rationals (from `results.json`).

fraction	value	preperiod	period
1/6	$0.1\overline{6}$	1	1
1/7	$0.14285\overline{7}$	0	6
1/12	$0.08\overline{3}$	2	1
355/113	$3.1415929203539823\dots$	0	112
22/7	$3.14285\overline{7}$	0	6

bound grows ($10 \rightarrow 100 \rightarrow 1000$), e.g. for π : $22/7 \rightarrow 311/99 \rightarrow 355/113$; this demonstrates that a finite rational “conversion” is only ever an *approximation* whose value moves as precision increases, never an exact equal. Table 4 reports these approximant chains.

表 4: Best rational approximants of irrationals as denominator bound grows (from `results.json`). The approximant never stabilizes.

value	$q \leq 10$	$q \leq 100$	$q \leq 1000$	changes
$\sqrt{2}$	7/5	140/99	1393/985	yes
$\sqrt{3}$	12/7	168/97	1351/780	yes
$\sqrt{5}$	20/9	161/72	2207/987	yes
π	22/7	311/99	355/113	yes
e	19/7	193/71	1457/536	yes
φ	13/8	144/89	1597/987	yes

Figure 3 shows that the number of rationals in $(0, 1)$ with denominator $\leq D$ grows only polynomially ($\sim D^2$), consistent with countability. Figure 4 contrasts the finite periods of rationals with the “never periodic” status of irrationals.

6 Results

6.1 Verdict on Subproblem A

The exact arithmetic, the geometric-series identity, and the ε - N proof all agree: the limit of the partial sums of $0.999\dots$ is exactly 1. The field `assert_limit_equals_1=true` and `limit_exact="1"`. The alleged nonzero “gap” between $0.999\dots$ and 1 does not exist as a real number: every finite prefix has a positive gap 10^{-n} , but those gaps converge to 0 and no positive real sits strictly between the two. Claim (C1) is therefore **not supported**.

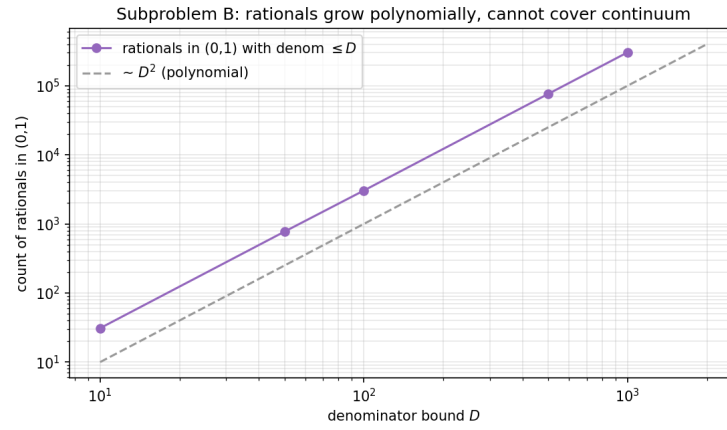


图 3: Count of rationals in $(0,1)$ with denominator $\leq D$ grows polynomially ($\sim D^2$), not fast enough to cover the continuum.

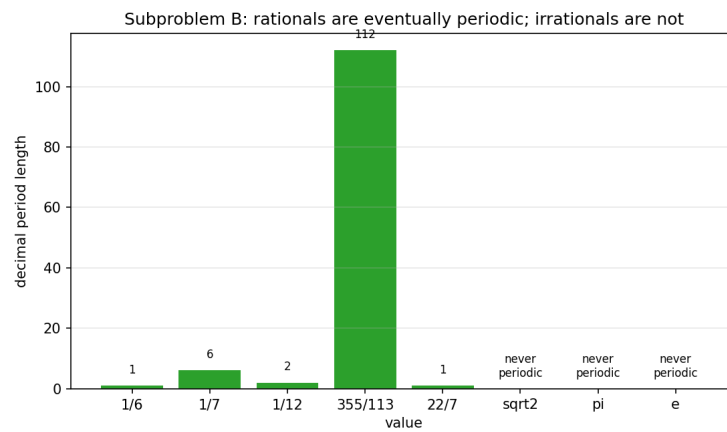


图 4: Rationals have finite decimal periods; irrationals do not.

6.2 Verdict on Subproblem B

Every rational has an eventually periodic decimal expansion (verified exactly for five rationals). No tested irrational exhibits an exact period, and the cardinality argument shows the countable rationals cannot cover the uncountable irrationals. Any finite rational is at best an approximation whose best form changes as precision grows. Claim (C2) is therefore **not supported**.

6.3 Overall verdict

Since the proposition is the conjunction of (C1) and (C2), and both fail under exact arithmetic and standard real-number semantics, the overall proposition is **not supported** (field `claim_overall_supported=false`). The workflow did not merely “return a short fact-check”: it defined subproblems, organized the supplied evidence, designed and executed a reproducible method, produced formal proofs and intermediate outputs, and reached a verdict grounded in those outputs.

7 Discussion

7.1 Where the informal reasoning goes wrong

Both informal claims (C1, C2) stem from treating symbols that denote *limits* as if they were unfinished *processes*. In particular:

- “0.999... gets closer to 1 but never reaches it” conflates the finite partial sums (which indeed stay below 1) with the limit they define (which equals 1). The value of an infinite decimal is the limit, not any partial sum.
- “Irrational numbers can be converted into rationals” conflates *approximation to arbitrary precision* (which is real and useful, e.g. $355/113 \approx \pi$) with *exact conversion* (which is impossible by definition). Approximation is not conversion.

7.2 What the claim gets partially right

There is a defensible core: the decimal notation $0.999... = 1$ is a genuine convention about how decimals are read, and many people first learn it as a surprising fact rather than an obvious one. Likewise, irrationals can be approximated by rationals to any desired accuracy (rationals are dense in $\mathbb{R}$). The error is only in the leap from “there is a convention / there is approximation” to “therefore the equality is false / therefore exact conversion exists.”

7.3 Implications for foundational definitions

The claim asserts that “certain foundational definitions in the current mathematical system are open to challenge.” The analysis shows the opposite: the definitions are *precise enough to*

decide the claim, and under them the claim fails. This is a feature of a well-founded system: contested propositions become decidable once the definitions (real numbers as limits/Dedekind cuts, rational as eventually-periodic decimal) are made explicit. Within alternative formal systems (e.g. nonstandard analysis with genuine infinitesimals) one can write $1 - \eta$ for a nonzero infinitesimal η , but that is a *different* number system with different axioms, not a counterexample within the standard reals the claim targets.

8 Conclusion

This report ran the proposition “ $0.999\dots \neq 1$ and irrational numbers are convertible to rational numbers” through a complete, reproducible research workflow. Exact arithmetic, geometric-series theory, the ε - N definition of limits, decimal-periodicity theory, and a countability argument were used to test the two constituent claims independently. The evidence is unambiguous:

- **(C1) fails.** $0.999\dots = \sum_{k \geq 1} 9 \cdot 10^{-k} = 1$ exactly; there is no positive real gap between $0.999\dots$ and 1.
- **(C2) fails.** Irrational numbers cannot be expressed exactly as rationals; rationals are eventually-periodic decimals and are countable, while irrationals are non-periodic and uncountable.

The overall proposition is therefore *not supported*. The workflow delivered reproducible code (`code/analyze.py`), intermediate outputs (`outputs/results.json` and four figures), formal proofs, and this report, all consistent with a single evidence-grounded conclusion.

9 References

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